



CRPF TenderGuard AI

Automating Government Procurement Evaluation

"Read every document. Check every criterion. Explain every verdict."

AI FOR BHARAT

THE PROBLEM



Manual & Slow

3-5 days per tender
600+ pages reviewed
by hand



Human Error

Two evaluators extract
different values from
the same document



Legal Risk

Wrongful disqualifications
CAG audit findings
RTI challenges

Result: Delayed procurement, unfair vendor treatment, public accountability failures

WHY IT MATTERS

"Government procurement evaluation is broken in a specific, fixable way."

10-20

Bids per
Tender

30-80

Documents
per Bidder

600+

Pages
Reviewed

2+

Languages
(Hindi & English)

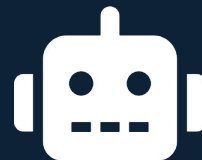
Impact: Delays, unfair vendor treatment, CAG audits, RTI challenges

OUR SOLUTION

An AI-powered end-to-end platform for government tender evaluation.

Officer reviews, AI extracts — not the other way around.

- ✓ System reads every document — no skipping
- ✓ AI checks every criterion — no misses
- ✓ Every decision has an exact source citation
- ✓ Ambiguous cases escalated to human review



TenderGuard AI

Tender Upload

AI Extraction

Bid Evaluation

Audit Report

Procurement officer signs off on a report — not a guess.

SYSTEM ARCHITECTURE

01

TENDER PARSING

- Extract eligibility criteria
- Classify: Financial / Experience
Technical / Compliance / Legal
- Build structured knowledge base

02

DOCUMENT INGESTION

- OCR for scanned docs & images
- Multi-language: Hindi + English
- Classify & extract fields
from bidder documents

03

EVALUATION & REPORT

- Match each bid to each criterion
- Verdict: Eligible / Not Eligible
/ Needs Review
- Source citations + audit trail

React 19

TypeScript

Firebase

Gemini AI

Vite

Node.js

Firestore

PHASE 1: TENDER PARSING

1

Upload tender document (PDF, image, DOCX, scanned)

2

AI reads and extracts all eligibility criteria automatically

3

Criteria classified by type and mandatory/optional status

4

Structured knowledge base created for evaluation engine

Criteria Types

Financial

Experience

Technical

Compliance

Legal

Output: Structured knowledge base of criteria — ready for automated matching

PHASE 2: DOCUMENT INGESTION

Accepted Document Formats

PDF

DOCX

JPEG/PNG

Scanned docs

OCR Engine

Converts scanned PDFs and images into machine-readable text
Supports Hindi (Devanagari) + English documents

Per Bidder Output

- ✓ Classified documents
- ✓ Extracted fields
- ✓ Parsed dates
- ✓ Verified values
- ✓ Cited sources

Output: Structured bidder profiles mapped to criteria — ready for evaluation

PHASE 3: EVALUATION ENGINE

For each bidder × each criterion:



Possible Verdicts

ELIGIBLE

All criteria met with cited evidence

NOT ELIGIBLE

Missing or invalid documents found

NEEDS REVIEW

Ambiguous — escalated to officer

Every decision includes exact page reference and quoted text from source document.

KEY FEATURES



Multi-Format Support

PDF, image, DOCX, scanned documents — all accepted



Audit Trail

Source citations for every single decision made



Multi-Language OCR

Hindi & English documents processed natively



Human-in-the-Loop

Ambiguous cases escalated to procurement officer



Criterion Extraction

Structured extraction of all eligibility criteria



Exportable Reports

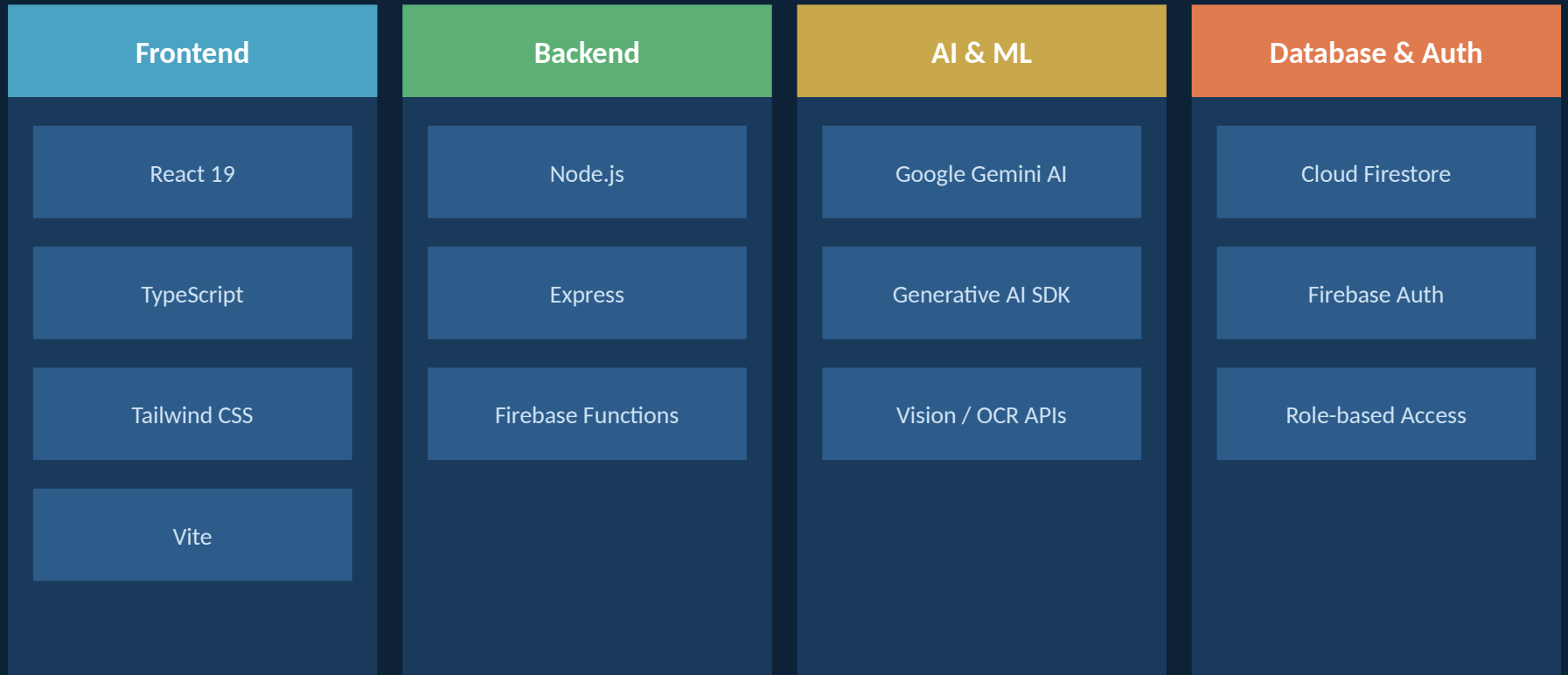
Full audit-ready report for each evaluation



Automated Evaluation

Every bid matched against every criterion

TECHNOLOGY STACK



Security: Role-based access control — Procurement Officer vs. Bidder roles enforced at all layers

IMPACT & OUTCOMES

< 4 hrs

Evaluation time
(was 3–5 days)

0

Manual extraction
errors

100%

Criteria audited
with citations



Speeds up procurement lifecycle from weeks to hours



Eliminates wrongful disqualifications via traceable decisions



RTI-challenge-proof with document-level audit trail



Scales across all government tenders — multiple languages, formats, complexities



Applicable to CRPF, defence, PSU, and all central procurement bodies

DEPLOYMENT & SCALABILITY

Deployment Model

- ✓ Web-based SaaS platform
- ✓ Government-cloud ready
- ✓ On-premise deployment option
- ✓ Multi-agency multi-tenant

Target Customers

-  CRPF & paramilitary forces
-  Defence procurement boards
-  PSUs & central ministries
-  State government agencies

Revenue Model

Annual SaaS license per agency · Per-tender usage pricing · Implementation & training services